Leveraging Transformer Models for Extractive Question Answering in Criminal Report Analysis

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***Abstract*—Background:Extractive question answering (QA) systems have gained significant attraction in various domains, including legal and criminal justice, due to their ability to automate information retrieval from large text corpora. Transformer-based models, such as BERT and its variants, have demonstrated remarkable performance in natural language processing (NLP) tasks, including QA. However, their application in domain-specific contexts, such as legal document analysis, requires careful adaptation to ensure accuracy and relevance. In the context of criminal justice, efficient extraction of information from crime reports is critical for case handling, resource allocation, and decision-making. This study focuses on robbery cases handled by the Fiscalía General del Estado in Ecuador, where the volume of reports and the complexity of legal language pose significant challenges for manual analysis. By leveraging transformer-based models, we aim to enhance the efficiency and accuracy of information retrieval from these reports, thereby supporting legal practitioners in their workflows and helps provide a better service for the community.**

Objective:The primary objective of this study is to investigate the effectiveness of fine-tuning transformer-based models for extractive QA tasks in the domain of criminal justice, specifically targeting robbery case reports. We aim to adapt a pre-trained transformer model to the unique linguistic and structural characteristics of legal documents, enabling it to accurately extract answers to common questions relevant to robbery cases. By doing so, we seek to demonstrate the potential of transformer models in optimizing information retrieval processes within the justice system, ultimately contributing to more efficient case handling and better resource allocation. The results of this research could be considered to improve current digital justice systems.

Methods:To achieve our objective, we employed a multi-step methodology. First, we prepared a dataset tailored to the domain of robbery case reports. Given the scarcity of annotated legal datasets, we utilized generative artificial intelligence (GAI) to generate synthetic question-answer pairs based on a sample of real robbery reports using Meta-Llama-3-8B-Instruct . This approach allowed us to create a diverse and representative dataset without compromising data privacy. The dataset was subsequently preprocessed and formatted according to the Stanford Question Answering Dataset (SQuAD) standards, specifically organizing the data into context, question and answer segments. Dataset generation utilized Huggingface Inference Client in order to save local computer power. The dataset was divided into training and test sets to evaluate the model's performance. A total of 4572 samples were used, with 3657 designated for training and the remaining 915 reserved for testing.

Next, we fine-tuned a Bert variation pre-trained transformer model (bert-base-spanish-wwm-cased-finetuned-spa-squad2-es) using the prepared dataset. The fine-tuning process involved optimizing hyperparameters, such as learning rate, batch size, and number of epochs, to ensure optimal performance on the extractive QA task. The training was performed using an NVIDIA Tesla T4 in Google Colab. The fine-tuned model was evaluated using standard metrics for QA models, including Exact Match (EM) and F1 score, to assess its performance in extracting relevant information from the crime reports.

Results:The fine-tuned transformer model demonstrated significant improvements in performance compared to the base pre-trained model. Specifically, the fine-tuned model achieved an F1 score of 82.88, surpassing the pre-trained model's F1 score of 81.70. This improvement indicates that the fine-tuning process successfully adapted the model to the domain-specific characteristics of robbery case reports, enabling it to extract more accurate and relevant information. Additionally, the model exhibited robust performance across a range of question types, including those requiring the identification of specific details such as dates, locations, and robbed objects.

Conclusion:This study demonstrates the effectiveness of fine-tuning transformer-based models for extractive QA tasks in the domain of criminal justice. By adapting a pre-trained model to the unique characteristics of robbery case reports, we achieved significant improvements in performance, as evidenced by the higher F1 score. The results underscore the potential of transformer models to optimize information retrieval processes within the justice system, enabling more efficient case handling and better resource allocation. This could help to improve current digital systems and provide in the near future a better service to the community by helping the prosecutors in their investigations and leveraging their workloads.

Furthermore, our use of generative artificial intelligence to create a synthetic dataset highlights a practical approach to overcoming the challenges of data scarcity and privacy concerns in legal domains. Future work could explore the application of this methodology to other types of legal documents, as well as the integration of additional NLP techniques, such as named entity recognition and relation extraction, to further enhance the capabilities of QA systems in legal analytics. Overall, this study contributes to the growing body of research on the application of AI in the justice system, paving the way for more innovative and impactful solutions.

***Index Terms*—transformer, natural language processing, law, machine learning**

I. Introduction

Transformer models have emerged as powerful tools in natural language processing, enabling more accurate and efficient extraction of relevant information from complex texts such as criminal reports. These models utilize self-attention mechanisms to weigh the importance of different words in a sentence, allowing for better context understanding and improved performance in tasks like question answering. By fine-tuning these models on domain-specific datasets, researchers can enhance their ability to discern pertinent details and provide precise answers to inquiries related to criminal incidents.

The implementation of extractive question answering systems in legal contexts not only enhances the efficiency of information retrieval but also addresses challenges related to data variability and complexity inherent in criminal reports. As these models are fine-tuned on diverse datasets, they can improve their generalizability across different types of legal documents, ensuring that critical details are accurately captured regardless of nuances in language or structure [2]. Moreover, recent advancements in model architectures, such as the integration of dynamic masking techniques, have shown promise in further refining performance by enabling models to better adapt to varying levels of difficulty in questions posed about complex narratives [4]. This evolution highlights the potential for transformer-based models to facilitate more informed decision-making processes within the justice system, ultimately contributing to enhanced outcomes for legal practitioners and stakeholders alike.

The Fiscalía General del Estado, which is the Prosecutor's Office in Ecuador, handles cases ranging from criminal investigations to public safety concerns, aiming to uphold justice and protect the rights of citizens. In this context, crimes of robbery are highly prevalent and present significant challenges for law enforcement agencies. Understanding the patterns and motivations behind these crimes can lead to more effective prevention strategies and better resource allocation, ultimately fostering a safer community for all citizens. Additionally, it is important to analyze all the information gathered through crime reports. Many users also rely on this information to access insurance coverage for items lost during theft (e.g., cell phones, computers, automobiles, automobile parts, etc.). The volume of cases that need to be reviewed by prosecutors is substantial. Consequently, it is of interest to explore the use of artificial intelligence in designing and training models that can help reduce the high demand and improve public service.

In previous research, the classification problem of different types of robbery was explored using transformer-based models and fine-tuning. This paper further investigates the capabilities of extractive question answering. This approach could help retrieve important information from criminal reports, which could be used in investigations conducted by the Prosecutor's Office as well as for statistical reports on robbery crimes.

This paper is organized as follows. In Section II, some common works in scientific literature regarding the use of Artificial Intelligence (AI) in Law is performend. Our methodology is presented in Section III, where we describe the construction of the dataset and the model training. In Section IV, the performance of the fine tuned model is presented. Finally, relevant conclusiones and future works are addressed in Section V

II. Literature Review

There is a significant interest within both the scientific community and legal institutions in studying the applications of AI in this expansive field. The current state of the art in Natural Language Processing (NLP) with Deep Learning (DL) primarily focuses on the use of transformer-based architectures, which have demonstrated superior performance across various NLP tasks.

In [7] the deployment of a RoBERTa Base question-answer classification model specifically for legal questions is discussed. It compares the performance of this transformer model against a traditional linear SVM in the legal domain, using the PRIVACYQA dataset. The results indicate that RoBERTa significantly outperforms the SVM, achieving a 31% improvement in F1-score and a 41% improvement in Mean Reciprocal Rank, demonstrating the effectiveness of transformers in enhancing answer retrieval for legal inquiries. In [6], the researchers focus on a Transformer-based neural network for answer selection in general question answering (QA) systems, not specifically in the legal domain. While Transformers are effective in extracting global information and improving answer selection, the research does not address their application in law-related question answering. For insights on using Transformers in legal contexts, further research specific to legal NLP applications would be necessary. For information retrieval, [3] discuss the use of transformer-based approaches for legal information retrieval and entailment, specifically in the context of the COLIEE competition. For case law retrieval (Task 1), a sentence-transformer model was employed to create numeric representations of case paragraphs, facilitating similarity assessments. In entailment (Task 4), a fine-tuned DeBERTa large language model was utilized, achieving third place among eight teams. These methods demonstrate the effectiveness of transformers in addressing legal question answering challenges. Another area of interest is to predict legal judgments. In [1] six transformer-based models (BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, and BigBird) are evaluated in predicting legal judgments using the ILDC dataset. It highlights the potential of these models in automating legal decision-making, with newer models achieving up to 80% accuracy compared to 76% for older models. This comparative analysis underscores the advancements in NLP technology and its applicability in the legal domain, suggesting a promising future for question answering in law through transformer models. Finally, in [5] the use of transformers to generate legal text is addressed, indicating their potential to assist in legal practice. The findings suggest that these models can effectively generate judicial opinions and help differentiate between human and machine-generated legal texts, which could be beneficial for tasks like drafting contracts and briefs, although direct application to question answering is not discussed.

III. Methodology

In this paper, we present the model `LeninGF/fge-robos-qa-model`[1], which is a fine-tuned version of mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-squad2-es. This model has been developed for the Fiscalía General del Estado to facilitate the extraction of answers from robbery crime cases. Our methodology is divided into

[1]LeninGF/fge-robos-qa-model

the following steps, which are explained in the subsequent sections:

1) Dataset Preparation
2) Prompt Engineering
3) Dataset Preprocessing
4) Model Fine Tuning

A. Dataset Preparation

In order to prepare the dataset for this task, Generative Artificial Intelligence (GAI) was used in order to extract the answers for a defined group of common questions in robbery cases. The model used in this case was `meta-llama/Meta-Llama-3-8B-Instruct`. First, an analysis of the number of words per crime report was performed. It showed a mean of 112 words with an upper limit of 150. Beyond 150, it could be considered outlier cases and for the lower limit of 82, cases with less words could also be rejected. As a consequence, we have considered to build the dataset only the crime report w whose number of words $\Gamma(w)$ is $82 \leq \Gamma(w) \leq 150$. Figures 1 and 2 show the histogram of $\Gamma(w)$ and a box Plot, respectively. The original dataset has a total of 174594 samples.

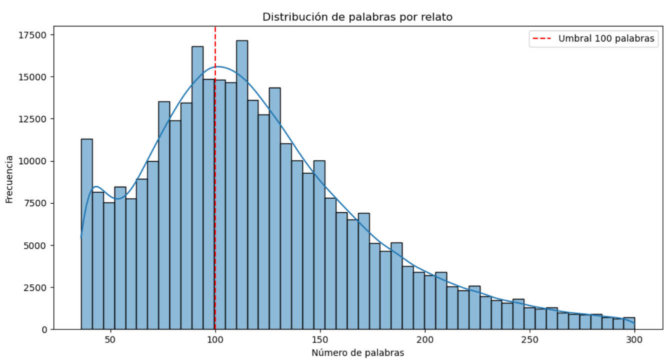

Fig. 1. Histogram Original Dataset

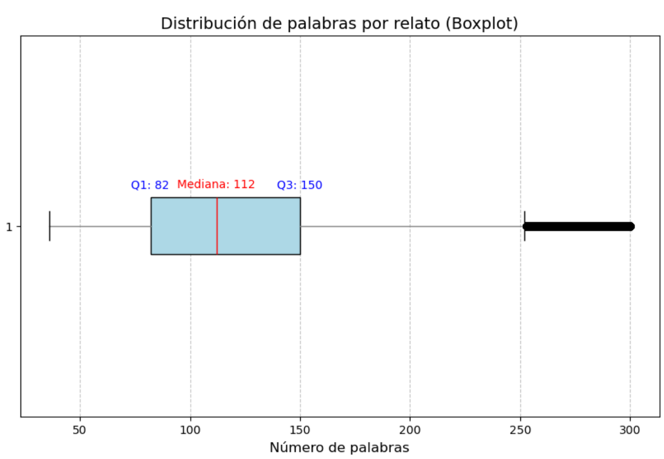

Fig. 2. Box Plot Original Data

B. Prompt Engineering

In this section, we describe the use of a Large Language Model (LLM) to extract answers to a set of common questions listed in Table I. The LLM functions as an expert data science assistant with the goal of constructing a dataset in the SQuAD format in Spanish. For this purpose, the prompt receives the content of the crime report as the `context` and each of the questions from Table I as the `question`. The LLM was also instructed to retrieve a `json` format output with the variables: `context`,~question~, `answer_text`, `answer_start`, `answer_end`, and `impossible_find_answer`. The latter where calculated by finding the `answer_text` in the context. In the case that no answer was possible to be retrieved, either because there is no answer or because the LLM failed, the variable `impossible_find_answer` was set to True.

TABLE I
COMMON QUESTIONS

Question
¿Qué objetos fueron robados?
¿En qué fecha ocurrió el incidente?
¿A qué hora sucedió el robo?
¿En qué dirección o entre qué calles sucedió el robo, suceso incidente?
¿Qué valor en dólares tenían los objetos sustraídos o robados?

After 4 hours of using the Inference Client from Hugging Face, a total of 1544 samples were processed, representing approximately 1% of the entire dataset. Due to computing constraints, it may be necessary to explore alternative approaches to optimize processing time and increase the number of samples processed. With the current approach, which allows for 20000 queries per day, it would take 43 days to scan the entire dataset. Additionally, upon reviewing the results, it was found that the LLM model failed to retrieve some answers and occasionally generated unrelated content. Consequently, manual verification of the obtained results is necessary. This procedures can be found in `dataset_build.ipynb` in our GitHub repository.

C. Dataset Preprocessing

Once we obtained the 1544 samples, we proceeded to check their quality. In this case the variable `impossible_find_answer` helped to locate cases where the LLM failed. They were around 3000 cases. It must be noticed that the total dataset extracted at this stage was of $1544 \times 5 = 7705$ samples. Because each context or crime report received 5 questions. As a consequence, we kept only the cases where `impossible_find_answer=False`. Next, the number of words was studied with the aid of an histogram and a box plot for the answers. It was found that the upper limit should be 17 and also that there are cases where the number of words in the answer is of 0. Figure 3 shows the histogram of the number of words in answer and Figure 4 shows the corresponding box plot. Consequently the conditions to accept samples were the following:

- $\Gamma(w) > 0$
- $impossible_{f}ind_{a}nswer = False$

Thus, a total of 4572 samples were used to form the final dataset for training and testing. The dataset was saved in LeninGF/robos-question-answering. Our code can be found at `dataset_xplore_and_upload.ipynb`

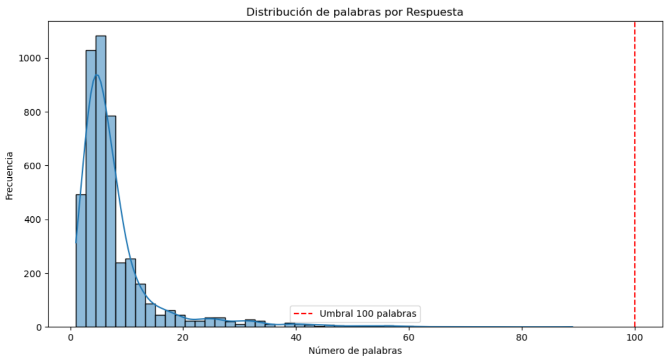

Fig. 3. Histogram Word Count Answers

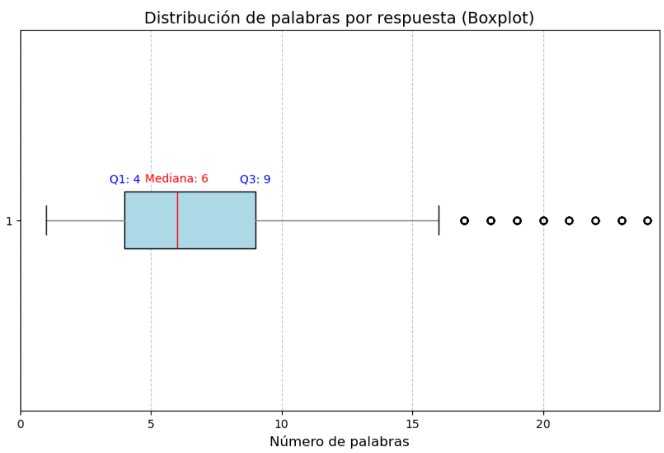

Fig. 4. Box Plot Word Count Answers

D. Model Fine-Tuning

In this section we described our methodology to train a question answering model for the crime reports of Robbery of the Fiscalía General del Estado del Ecuador. The code can be found in ~question-answering-Bert.ipynb~The procedure was the following

1) Dataset Train and Test Split: First, the dataset was divided into training and testing sets, with 20% of the cases allocated for performance evaluation. Table II displays the number of samples in each set

TABLE II
DATASET TRAIN AND TEST SPLIT

Partition	# Samples
Train	3657
Test	915
Total	4572

2) Pre-trained Model Loading: The `mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-` was selected to be fine tuned because it is a model trained for Spanish and Question Answering Tasks. As a matter of fact, it was tested without training and it performed with an F1-Score of 81.70.

3) Preprocessing: To fine-tune the model, both the question and the answer need to be tokenized. As a result of this process, the indices indicating the start and end of the answer must be recalculated based on the tokenized embedding. In the case of long text, truncation should be applied only to the context, and a window of 128 tokens was used to avoid losing the answer during truncation. However, in this scenario, since most of the current samples have no more than 100 words in the context, this issue is unlikely to occur; nonetheless, it has been addressed in the program.

4) Evaluation Metrics: Our next step was to define the metrics to be used to evaluate the performance of the model. In this case we use `squad_v2` which contains both the F1-SCORE and EXACT MATCH. A function was prepared in order to be used by the Trainer Class.

5) Training: In order to train the model, the parameters shown in Table III were used. A small learning rate is recommended in Fine Tuning in order to re-use the pre-training from the original model and adapt the new model to the target dataset. Our results prove that it is possible to achieve a better performance on the target domain with the fine tuned model rather than using the pre-trained model. Although the difference in performance is small, it could be related to the fact that we were not able to use the whole dataset.

TABLE III
TRAINING PARAMETERS

Parameter	Value2
Strategy	epoch
Learning rate	2×10^{-5}
batch	64
epochs	10
weight decay	0.01
fp16	True
metric for best model	F1 (eval)

IV. RESULTS

In this section, we present the results of fine-tuning the model, along with the performance of the pre-trained model on the test set. This comparison was conducted to determine whether the fine-tuned model demonstrated any improvements over the original pre-trained model. Table IV shows the performance of the pre-trained model while Table V shows the performance of the fine-tuned model. It is worth to notice that the latter improves over the former. The fine-tuned model achieved an F1 SCORE of 82.88 on evaluation dataset.

As an example, we provide the following text and the answers that our fine-tuned model was able to extract from it in Table VI together with an score.

> el día de ayer 17 de junio del 2020 aproximadamente a las 10h00 estaba realizando una factura en el sector de cotocollao av legarda instantes en los cuales fui

TABLE IV
PRE-TRAINED MODEL PERFORMANCE

Metric	Value
eval_loss	0.9102
eval_model_preparation_time	0.0077
eval_exact	53.5519
eval_f1	81.7041
eval_total	915.0000
eval_HasAns_exact	53.5519
eval_HasAns_f1	81.7041
eval_HasAns_total	915.0000
eval_best_exact	53.5519
eval_best_exact_thresh	0.0000
eval_best_f1	81.7041
eval_best_f1_thresh	0.0000
eval_runtime	12.3982
eval_samples_per_second	73.8010
eval_steps_per_second	1.2100

TABLE V
FINE TUNED MODEL PERFORMANCE

Metric	Value
eval_loss	1.0088
eval_model_preparation_time	0.0077
eval_exact	55.7377
eval_f1	82.8805
eval_total	915.0000
eval_HasAns_exact	55.7377
eval_HasAns_f1	82.8805
eval_HasAns_total	915.0000
eval_best_exact	55.7377
eval_best_exact_thresh	0.0000
eval_best_f1	82.8805
eval_best_f1_thresh	0.0000
eval_runtime	12.8109
eval_samples_per_second	71.4240
eval_steps_per_second	1.1710
epoch	10.0000

sorprendido por un sujeto de acento costeño mismos que bajo amenazas y con una cuchillo procedido a sustraerme lo siguiente un celular marca samsun modelo a10 s duo color negro de la operadora movistar asignado con el número xxxxxxxxxx imei xxxxxxxxxx perteneciente a la empresa alimentos yupi con lo antes expuesto solicito las respectivas investigaciones para mayor información comunicarse a los números xxxxxxxxxx

TABLE VI
EXAMPLE OF QUESTION ANSWER EXTRACTION

Question	Answer	Score
¿Qué objetos fueron robados?	un celular marca samsun modelo a10 s duo color negro	0.966
¿En qué fecha ocurrió el incidente?	el día de ayer 17 de junio del 2020	0.615
¿A qué hora sucedió el robo?	aproximadamente a las 10h00	0.983
¿En qué dirección o entre qué calles sucedió el robo, suceso o incidente?	sector de cotocollao av legarda	0.483
¿Qué valor en dólares tenían los objetos sustraídos o robados?	10 7234734	0.167

It can be noticed from the sample that the score the model provides is indicative. The crime report has no information regarding the value of the objects or any amount of money stolen. However, the answer has a low score. This can help to process further the extraction performed by the model.

V. CONCLUSION AND FUTURE WORKS

In this study, we demonstrated the effectiveness of transformer-based models for extractive question answering within the context of criminal report analysis. Our fine-tuned model showed a notable improvement, achieving an F1 score of 82.88 compared to the pre-trained model's score of 81.70. This indicates the potential for enhanced performance through domain-specific fine-tuning.

The application of our model to robbery crime reports could facilitate the extraction of crucial information, supporting legal practitioners in their investigations and reporting. However, the study also highlighted some limitations, including the model's occasional inability to retrieve accurate answers and the need for manual verification of results.

Future work will focus on addressing these limitations by exploring other advanced techniques and by manual labeling of the dataset by experts peers. Additional computational resources are also necessary to process large datasets. Moreover, the integration of artificial intelligence in the legal domain holds promise for more efficient case handling and resource allocation, ultimately contributing to a safer community.

We encourage further research and collaboration to refine these approaches and harness the full potential of AI in supporting the justice system.

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